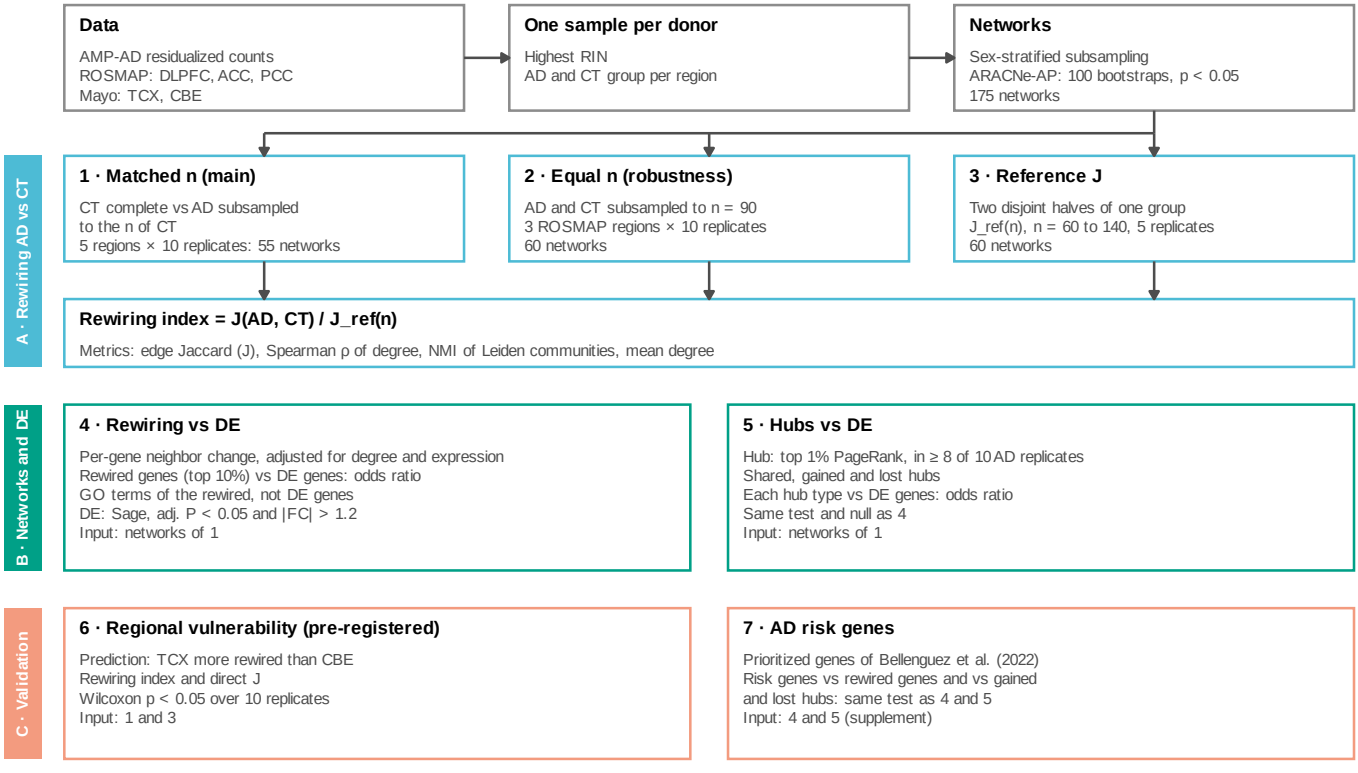
Data
AMP-AD residualized counts
ROSMAP: DLPFC, ACC, PCC
Mayo: TCX, CBE
One sample per donor
Highest RIN
AD and CT group per region
Networks
Sex-stratified subsampling
ARACNe-AP: 100 bootstraps, p < 0.05
175 networks
A · Rewiring AD vs CT
1 · Matched n (main)
CT complete vs AD subsampled
to the n of CT
5 regions × 10 replicates: 55 networks
2 · Equal n (robustness)
AD and CT subsampled to n = 90
3 ROSMAP regions × 10 replicates
60 networks
3 · Reference J
Two disjoint halves of one group
J_ref(n), n = 60 to 140, 5 replicates
60 networks
Rewiring index = J(AD, CT) / J_ref(n)
Metrics: edge Jaccard (J), Spearman ρ of degree, NMI of Leiden communities, mean degree
B · Networks and DE
4 · Rewiring vs DE
Per-gene neighbor change, adjusted for degree and expression
Rewired genes (top 10%) vs DE genes: odds ratio
GO terms of the rewired, not DE genes
DE: Sage, adj. P < 0.05 and |FC| > 1.2
Input: networks of 1
5 · Hubs vs DE
Hub: top 1% PageRank, in ≥ 8 of 10 AD replicates
Shared, gained and lost hubs
Each hub type vs DE genes: odds ratio
Same test and null as 4
Input: networks of 1
C · Validation
6 · Regional vulnerability (pre-registered)
Prediction: TCX more rewired than CBE
Rewiring index and direct J
Wilcoxon p < 0.05 over 10 replicates
Input: 1 and 3
7 · AD risk genes
Prioritized genes of Bellenguez et al. (2022)
Risk genes vs rewired genes and vs gained
and lost hubs: same test as 4 and 5
Input: 4 and 5 (supplement)